

Technical Document #121: Data Engineering - Comprehensive Overview

Technical Document #121: Data Engineering - Comprehensive Overview

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Data governance establishes policies and procedures for managing data assets.
- Data lineage tracks the origin and movement of data through systems.
- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a data warehouse or lake.